

torch.nn.attention.bias.causal_upper_left

https://pytorch.org/docs/stable/generated/torch.nn.attention.bias.causal_upper_left

Description:

Creates an upper-left triangular causal bias. This function generates a upper-left triangular matrix to represent causal attention bias with a diagonal offset set so that the inclusive values are aligned to the upper left corner of the matrix. This equivalent to the `is_causal=True` argument in `scaled_dot_product_attention`. The equivalent pytorch code for constructing this bias is: For instance, with `shape=(3,4)`, the materialized bias tensor will be: `size - The size of the bias matrix.`

Function Signature

```
torch.nn.attention.bias.causal_upper_left(*size)[source]#
```

Parameters

Returns

Return type

Returns:

CausalBias

Code Examples

```
torch.tril(torch.ones(size, dtype=torch.bool))
```

```
[[1, 0, 0, 0],  
 [1, 1, 0, 0],  
 [1, 1, 1, 0]]
```

Detailed Documentation

Documentation

Creates an upper-left triangular causal bias.

This function generates a upper-left triangular matrix to represent causal attention bias with a diagonal offset set so that the inclusive values are aligned to the upper left corner of the matrix. This equivalent to the `is_causal=True` argument in `scaled_dot_product_attention`.

The equivalent pytorch code for constructing this bias is:

For instance, with `shape=(3,4)`, the materialized bias tensor will be:

size – The size of the bias matrix.

The UPPER_LEFT triangular causal bias variant.